

Understanding a New Medical Diagnostic

Test:

The TriOx Blood Test for Early Cancer

Detection

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ITAI 4375- AI in Healthcare-RT-17185

Spring 2026

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February 16, 2025.

Introduction

Scientists at the University of Oxford have recently developed an exciting new blood test called TriOx that could change the way we detect cancer. This report will explain what this test does, how well it works, and who might benefit from using it. The information comes from a recent study published in January 2025 by the university (University of Oxford, 2025).

What Is the TriOx Test For?

The TriOx test is a blood test designed to detect cancer in its earliest stages, when it is hardest to detect but easiest treatable stage. At present, the test can identify six different types of cancer (University of Oxford, 2025):

- Breast cancer
- Colorectal cancer (colon cancer)
- Esophageal cancer (cancer of the food pipe)
- Pancreatic cancer
- Ovarian cancer
- kidney cancer

What makes this test scientific breakthrough is that it is a 'liquid biopsy,' which means doctors only need to take a small blood sample instead of removing tissue from your body. The test assesses tiny pieces of DNA floating in your blood that come from cancer cells. These pieces are called circulating tumor DNA, or ctDNA for short (Vavoulis et al., 2025).

The test uses advanced technology called TAPS (which stands for TET-Assisted Pyridine Borane Sequencing) combined with machine learning, a type of artificial intelligence. This combination allows the test to assess many different features of the DNA all at once, making it much better at catching cancer than older tests that only look at one or two things (University of Oxford, 2025).

How Well Does the Test Work?

The researchers tested TriOx on blood samples from people who might have cancer because they had symptoms, as well as on healthy people without symptoms. The results were very promising. The test correctly identified cancer in 94.9% of people who had cancer. It also correctly identified people without cancer 88.8% of the time (University of Oxford, 2025).

To understand what these numbers mean, we need to know about sensitivity and specificity.

Understanding Sensitivity and Specificity

Sensitivity is how good a test is at finding people who have the disease. It answers the question: 'If someone has cancer, will the test catch it?' A sensitivity of 94.9% means that out of 100 people with cancer, the test will correctly identify about 95 of them. This is excellent because it means the test rarely misses cancer when it is there.

Specificity is how good a test is at correctly identifying people who do not have the disease. It answers the question: 'If someone doesn't have cancer, will the test correctly say they are healthy?' A specificity of 88.8% means that out of 100 healthy people, the test will correctly identify about 89 of them as not having cancer.

Both numbers are important. If sensitivity is too low, the test will miss people who have cancer. If specificity is too low, too many healthy people will be told they might have cancer, which causes unnecessary worry and extra testing (Vavoulis et al., 2025).

Can the Test Be Wrong?

Absolutely, like all medical tests, TriOx can sometimes give incorrect results. Based on the numbers above, we can understand two ways the test might be wrong:

False Negatives: About 5 out of every 100 people who have cancer might be told they do not have it (because the sensitivity is 94.9%, not 100%). This is concerning because these people might delay getting treatment they need.

False Positives: About 11 out of every 100 healthy people might be told they could have cancer when they do not (because the specificity is 88.8%). These people would need more testing to confirm they are healthy, which can cause stress and worry.

However, compared to many other tests, these error rates are relatively low. Professor Anna Schuh, who led the study, emphasizes that minimizing false results is very important because 'patients without cancer are not subjected to unnecessary procedures, while those with cancer can receive treatment sooner' (University of Oxford, 2025, para. 9).

Who Might Need This Test and Why?

Presently, the TriOx test is still being developed and is not yet available to the public. However, once it is fully ready, several groups of people might benefit from it:

People with symptoms: If someone goes to their doctor with symptoms that could be cancer, like unexplained weight loss, persistent pain, or unusual lumps, this test could help quickly figure out if cancer is the cause.

People at high risk: Those with a family history of cancer or other risk factors might use this test for regular screening to catch cancer as early as possible.

People want routine screening: In the future, researchers hope this test could become as common as cholesterol or blood sugar tests. People could get it during regular checkups to monitor for cancer before any symptoms appear (University of Oxford, 2025).

This test is particularly important for cancers like pancreatic and ovarian cancer, which Dr. Dimitris Vavoulis, a co-lead researcher, notes 'often go unnoticed until they have advanced, when treatment is more difficult and less effective' (University of Oxford, 2025, para. 10). Current screening methods cannot catch these cancers early, so TriOx could be lifesaving for people at risk.

My Opinion: Is This Test Good?

I think the TriOx test is very promising and represents an important step forward in cancer detection. Here is why I believe it is a good test:

First, the test is minimally invasive. Getting a blood test is much easier and less scary than other cancer detection methods like biopsies, where doctors must remove tissue from your body. This means more people might be willing to get tested regularly.

Second, the accuracy rates are impressive. A sensitivity of 94.9% means the test catches almost all cancers, which is crucial for saving lives. While the specificity of 88.8% means some healthy people will need follow-up tests, this is a reasonable trade-off to make sure cancer is not missed.

Third, the test can detect multiple types of cancer at once. Instead of needing separate tests for breast cancer, colon cancer, and so on, one blood test can screen for six different cancers. This is more convenient and could catch cancers that people were not specifically looking for.

Fourth, early detection saves lives. Cancer is much easier to treat when caught early. As Professor Schuh explains, this test could 'improve survival rates for millions worldwide' by catching cancer 'when it is easier to treat' (University of Oxford, 2025, para. 8).

However, there are some limitations to consider. The test is still in development, so we need to see how well it works in larger groups of people and with more cancer types. Also, the 11% false positive rate means some people will have to go through additional testing and worry unnecessarily. Finally, we do not yet know how much this test will cost or whether insurance will cover it.

Despite these concerns, I believe the benefits outweigh the drawbacks. The researchers are working to improve the test and expand it to detect even more cancers. If TriOx becomes widely available, it could fundamentally change how we approach cancer screening, making it routine, easy, and effective. This could save countless lives by catching cancer before symptoms even appear.

Conclusion

The TriOx blood test represents an exciting development in cancer detection. By using advanced DNA analysis and machine learning, it can detect six types of cancer with high accuracy using just a simple blood draw. With a sensitivity of 94.9% and specificity of 88.8%, it performs well at both finding cancer and avoiding false alarms. While the test is still being perfected, it shows great promise for making cancer screening more accessible and effective. People with symptoms, those at high risk, and eventually anyone wanting routine screening could benefit from this innovation. As research continues, tests like TriOx could make early cancer detection as common as any other routine blood test, potentially saving millions of lives.

References

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Oxford researchers develop blood test to enable early detection of multiple cancers

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Nature

Communications

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